Fiscal Year:	FY 2017	**Task Last Updated:**	FY 08/23/2016
PI Name:	Chahine, Georges D.Sc.		
Project Title:	Two-Phase Flow Separation and Characterization for Microgravity Applications		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	glchahine@dynaflow-inc.com	**Fax:**	FY
PI Organization Type:	INDUSTRY	**Phone:**	301-604-3688
Organization Name:	Dynaflow, Inc.		
PI Address 1:	10621 Iron Bridge Rd, Suite J		
PI Address 2:			
PI Web Page:			
City:	Jessup	**State:**	MD
Zip Code:	20794	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Fluid Physics NNH08ZTT002N
Start Date:	10/21/2015	**End Date:**	10/20/2020
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:	McQuillen, John	**Contact Phone:**	216-433-2876
Contact Email:	john.b.mcquillen@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Wu, Xiongjun (Dynaflow, Inc.) Loraine, Greg Ph.D. (Dynaflow, Inc.)		
Grant/Contract No.:	NNX16AB85A		
Performance Goal No.:			
Performance Goal Text:			
	NOTE: Continuation of "Microgravity Fluid Separation: Dynaswirl® Bubble Vortex Separator," grant NNX11AO76A with same Principal Investigator Georges Chahine, D.Sc. (Ed., 4/18/2016) The limits on the amount of liquids and gases carried to space make it imperative to recycle and reuse. On Earth, bubbles are relatively easily separable by buoyancy. In microgravity, other external forces such as a centrifugal force must be utilized to separate bubbles from liquids. Two categories of centrifugal force separators exist: one uses active rotation by mechanically spinning the tank. This is very efficient but requires shaft, bearings, and motor. In the second, the tank is fixed and the rotation is induced by eccentric injection of the mixture (Free Vortex Separator or FVS). This FVS has been proven very efficient for mid-range void fraction (50-80% gas) with an efficiency approaching 100%. However, this FVS is not as efficient for lower void fraction, and there is a need of a separator for low void fractions.		

Task Description:	We have developed a phase separator at DYNAFLOW based on a DYNASWIRL® swirl flow nozzle. It consists of two concentric cylinders, the fluid is introduced in the space between the two cylinders, and the swirling flow inside the inner cylinder is generated with the help of tangential injection slots. With this configuration one can make the vortex circulation intense enough to generate cavitation with a very stable vaporous/gaseous core even at low flow rates. Operation with the high swirl near or slightly below cavitation inception in the vortex core guarantees removal of air even from mixtures with very low void fractions. This is achieved by forcing the bubbles to expand and collect in the vortex core. These are then removed through two central exit orifices at each end of the cylinder. This configuration was tested successfully during a reduced-gravity flight and we are planning additional testing. We are further developing and experimentally testing this configuration to characterize its gas-liquid separation performance and to improve its efficiency. We are also conducting numerical simulations to help optimize the design and ultimately build a compact system for space validation on the International Space Station (ISS) using the existing Fluid Integrated Rack facility.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	As a passive separator device, the DYNASWIRL® phase separator has the advantage of simplicity and dependability. It requires no moving mechanical parts and low power. Compared to other passive separator devices, which usually are not very efficient at low void fraction, the DYNASWIRL® is able to force low void fraction gas out of the liquid into the central core of the vortex by combining the effects of swirl, cavitation, and rectified gas diffusion. In addition to being an effective phase separator for space applications, it will also find wide applications in Earth applications where fast separation of low void fraction gas is of concern.
Task Progress:	During this reporting period, we have focused our efforts on designing, constructing, and testing a new single wall DYNASWIRL® phase separator to improve the optical quality of the test section to enable good imaging while evaluating the performance of the phase separator. Systematic tests were conducted to evaluate the performance of the gas extraction tubes manufactured using different manufacturing processes, different geometric dimensions, and different membrane materials. On-going ground based experiments were also conducted to study the flow and pressure fields of the new test section under different flow conditions. This is combined with parallel numerical simulations. Additionally, we are actively working with the team from NASA Glenn Research Center (GRC) and Zin Technologies to finalize the SRD (science review document) and the test matrix with the tests on International Space Station (ISS) in mind. Several iterations of these documents have been made. A. Design of a Single-Wall Test Section: Based on previous ground based experiments as well as on reduced gravity flight tests, it was noted that the image quality from high-speed photography needs improvements for high magnification conditions. During this reporting period, we have designed and fabricated a new DYNASWIRL® phase separator with a single-wall design in order to eliminate optical distortions resulting from the earlier double cylindrical wall design. The newly constructed test section has a footprint of 8.875 in. x 7.5 in. x 7.5 in. The liquid exit chamber with two ABS ACOUSTIC BUBBLE SPECTROMETER®© hydrophones was also redesigned. A square section was adopted to improve viewing with the hydrophones flush mounted in two side walls. The new design eliminated flow recirculation which may interfere with the bubble size measurement. The new single-wall test section has been integrated into the test loop and is currently under extensive testing for performance evaluation. B. Improvement of Gas Extraction Scheme: In order to significantly improve gas extraction rates and prevent the gas from escaping into the liquid orifices, the following design changes were implemented: (a) The gas extraction tube length was increased from 6 inches to 8 inches ; (b) Several gas permeable membranes with different pore sizes, thickness, and materials were tested ; (c) The methods of construction of the tubes were investigated ; (d) High speed videos were taken of the bubble captured by the vapor core. The greater length increased the surface area of the extraction tube from 2.4 in2 to 3.1 in2. A new membrane type with an average pore size of 0.8 µm replaced the previous 0.45 µm pore size membrane. The new membrane demonstrated higher air flux rates and lower breakthrough water flow rates than the previous membrane. High speed video data was collected to visualize bubble behavior around the gaseous core and capture of individual bubbles by the core. The high speed imaging data was also be used to measure the velocity of the bubbles as a function of the radial distance from the vapor core based on particle/bubble tracking velocimetry. C. Numerical Simulations: 3D Numerical simulations were conducted by coupling our two-phase flow Navier-Stokes CODE 3DYNAFS-VIS©, which models the overall flow field of the bubbly mixture in the DYNASWIRL® phase separator, and 3DYNAFS-DSM©, a bubble tracking and bubble dynamics motion code. A bubble removal scheme was implemented in 3DYNAFS-DSM© to simulated gas extraction: the scheme removes the bubbles from the computational domain once they reach a radial position corresponding to the radius of the central gas extraction tube. This scheme allows us to quantify the gas extraction rate in terms of axial location. With such information the length of swirl chamber and gas extraction tube can be optimized. We also studied with the same software gravity effects on gas removal rates. D. Documentation Updates: As a new focus of the NASA team project is to prepare a flight-ready separator to be tested on ISS, members from Zin Technologies are now involved in assisting with the engineering work needed to deploy the separator on ISS. We have been working with the team members from NASA GRC and Zin Technologies to finalize the SRD needed and defining the test matrix for the planned tests on ISS. So far, several iterations of these documents have been made.
Bibliography Type:	Description: (Last Updated: 09/04/2025)
Papers from Meeting Proceedings	Wu X, Loraine G, Hsiao C-T, Chahine G. "Development of an Efficient Phase Separator for Space and Ground Applications." ASME 2016 Fluids Engineering Division Summer Meeting, FEDSM2016, Washington, DC, July 10-14, 2016. Proceedings of the ASME 2016 Fluids Engineering Division Summer Meeting, FEDSM2016, Washington, DC, July 10-14, 2016. Paper FEDSM2016-7793. , Jul-2016

Papers from Meeting Proceedings	Hsiao C-T, Ma J, Chahine G. "Numerical Study of Gravity Effects on Phase Separation in a Swirl Chamber." ASME 2016 Fluids Engineering Division Summer Meeting, FEDSM2016, Washington, DC, July 10-14, 2016. Proceedings of the ASME 2016 Fluids Engineering Division Summer Meeting, FEDSM2016, Washington, DC, July 10-14, 2016. Paper FEDSM2016-7845. , Jul-2016